

ClosureWatch

Restaurant Failure Prediction - Project Report

ML Final Project | Tampa Bay, FL | Yelp Academic Dataset

1. Project Overview

ClosureWatch predicts whether a Tampa Bay restaurant will permanently close within 6 months, using 12 months of behavioral signals from the Yelp Academic Dataset. The model is framed as a binary classification problem aimed at SME credit underwriting (e.g. Kabbage / BlueVine) - predicting business failure from public behavioral data rather than private financials.

- Dataset: Yelp Academic Dataset (public, not Kaggle)
- Geography: Tampa Bay metro area (then all of Florida if < 200 closures)
- Label: closed_within_6m (binary, 6.3% positive rate)
- 5,143 restaurants | 326 closures | Test set: 1,244 businesses

2. Pipeline - Run Order

All scripts must be run in numerical order. Each depends on the previous output.

```
python 01_load_filter.py          # Load Yelp JSON -> filtered Parquet (~5-10 min)
python 02_build_labels.py         # Anchor dates + leakage-safe binary labels
python 03_feature_engineering.py  # Time-windowed features (VADER, velocity, trend)
python 04_eda.py                  # EDA plots -> figures/
python 05_modeling.py             # 5 models with CV + test evaluation
python 06_ensemble.py            # Ensemble strategy ladder (NEW)
streamlit run app.py              # Launch interactive dashboard (NEW)
```

3. Temporal Design - Anti-Leakage Rules

The most critical correctness constraint. Each restaurant gets an 'anchor date' (its 80th-percentile review date). Features use only data before that date; the label is determined by what happens in the 6 months after.

- Features use ONLY data where date < anchor_date
- Anchor = 80th-pct review date (not last review - avoids trivial leak)
- Imputation medians computed on training fold only
- Train/test split is time-based (most recent 15% by anchor date = test)
- Cross-validation uses expanding-window folds, not random k-fold

4. Feature Engineering (03_feature_engineering.py)

Top predictive signals extracted from Yelp behavioral data:

Feature	Correlation	Description
months_with_zero_reviews	0.56	Months in obs. window with no reviews
days_since_last_review	0.56	Days from last review to anchor date
review_drought_flag	0.43	Binary: drought threshold exceeded
checkin_drought_flag	0.34	Binary: check-in drought triggered

5. Models Trained (05_modeling.py)

Five classifiers trained on the time-based train split and evaluated on the held-out test set. Primary metric: AUC-PR (area under precision-recall curve), chosen because the dataset is highly imbalanced (6.3% positive).

Model	CV AUC-PR	Test AUC-PR	AUC-ROC	Test F1
XGBoost (winner)	0.124 ± 0.039	0.2069	0.7002	0.2448
Random Forest	0.112 ± 0.054	0.1994	0.6948	0.1347
LightGBM	0.091 ± 0.021	0.1977	0.6699	0.2093
MLP Neural Net	0.083 ± 0.010	0.1592	0.6187	0.1622
Logistic Regr.	0.106 ± 0.035	0.1574	0.6458	0.2553

Base rate (random AUC-PR): 0.063 | XGBoost is 3.3x better than random

6. Ensemble Strategy (06_ensemble.py) - NEW

A strategy ladder that tries three approaches in order and stops at the first one that beats the XGBoost baseline (AUC-PR = 0.2069).

- Strategy 1 - Simple Average: equal-weight average of all model probabilities
- Strategy 2 - Weighted Average: weighted by CV AUC-PR score per model
- Strategy 3 - Stacking: meta-LogisticRegression trained on OOF predictions

The script is resilient to sklearn version mismatches: if a model's `predict_proba()` fails (e.g. due to a version bump), it is skipped with a warning and the ensemble continues with the remaining models.

Outputs:

- `models/ensemble_results.json` - metrics for each strategy + winner
- `data/processed/ensemble_predictions.parquet` - risk scores per restaurant

7. Streamlit Dashboard (app.py) - NEW

An interactive dark-themed dashboard built with Streamlit + Plotly. Launch with: `streamlit run app.py`

- Sidebar: all Tampa Bay restaurants sorted by predicted closure risk (%)
- Color-coded badges: RED $\geq$ 60% | ORANGE 30-59% | GREEN $<$ 30%
- Search bar: case-insensitive filter by restaurant name
- Detail panel: risk gauge, key warning signals, Plotly feature bar chart
- Feature bar chart: percentile-normalized contributions (red = risk-increasing, green = risk-decreasing)
- Data source: `data/processed/ensemble_predictions.parquet`

8. File Structure

Desktop/Final ML/	
<code>config_00.py</code>	Global constants (paths, dates, hyperparams)
<code>01_load_filter.py</code>	Yelp JSON -> filtered Parquet
<code>02_build_labels.py</code>	Anchor dates + binary labels
<code>03_feature_engineering.py</code>	Time-windowed feature extraction
<code>04_eda.py</code>	EDA plots
<code>05_modeling.py</code>	5 models, CV, test evaluation
<code>06_ensemble.py</code>	Ensemble strategy ladder [NEW]
<code>app.py</code>	Streamlit dashboard [NEW]
<code>generate_report.py</code>	This report generator [NEW]
<code>PROJECT_REPORT.pdf</code>	This document [NEW]

requirements.txt	Python dependencies
CLAUDE.md	Project guide for Claude Code
.streamlit/config.toml	Dark theme config
data/raw/	Yelp source JSON files (not committed)
data/processed/	Auto-generated Parquet files
models/	Saved .pkl files + results JSON
figures/	EDA plots (auto-generated)
tests/	pytest unit tests
test_ensemble.py	Tests for 06_ensemble.py math functions
test_app_helpers.py	Tests for app.py helper functions
docs/superpowers/	
specs/	2026-04-16-restaurant-closure-ui-ensemble-design.md
plans/	2026-04-16-ensemble-and-ui.md

9. Unit Tests

Run with: `pytest tests/ -v`

- test_ensemble.py (8 tests): simple_average, weighted_average, compute_metrics
- test_app_helpers.py (14 tests): risk_color, risk_label, risk_badge, percentile_rank

10. Quick-Start Guide

Prerequisites: Anaconda Python environment with the packages in requirements.txt

```
# 1. Place Yelp JSON files in data/raw/
# 2. Run the full pipeline
python 01_load_filter.py
python 02_build_labels.py
python 03_feature_engineering.py
python 04_eda.py
python 05_modeling.py
python 06_ensemble.py

# 3. Launch the dashboard
streamlit run app.py

# 4. Run tests
pytest tests/ -v
```